Exam 4. Practice 1.

*You may use a basic calculator and formula sheets. Your sheets must contain formulas only, no examples. Phones and all other smart devices are forbidden. Any AI usage will result in an automatic F for the exam. Show **ALL** your work for credits.*

1. Find $f'(x)$. Simplify 1c.
   a. $f(x) = 3x^3 - 7x + 9$

   $f'(x) = 9x^2 - 7$

   b. $f(x) = \frac{3}{x^6} + \frac{4}{x^7} + \frac{5}{x^9} + 3x + 1$

   $f(x) = 3x^{-6} + 4 \cdot x^{-7} + 5x^{-9} + 3x + 1$

   $\Rightarrow f'(x) = 3 \cdot (-6) \cdot x^{-6-1} + 4 \cdot (-7) x^{-7-1} + 5 \cdot (-9) x^{-9-1} + 3$

   $= -18x^{-7} - 28x^{-8} - 45x^{-10} + 3$

   c. $f(x) = \frac{x^2+1}{x^2-1}$

   $f'(x) = \frac{(x^2+1)' \cdot (x^2-1) - (x^2+1) \cdot (x^2-1)'}{(x^2-1)^2}$

   $= \frac{2x(x^2-1) - (x^2+1) \cdot 2x}{(x^2-1)^2}$

   $= \frac{2x^3 - 2x - 2x^3 - 2x}{(x^2-1)^2}$

   $= \frac{-4x}{(x^2-1)^2}$

   d. $f(x) = (x^3 + 1)(2x^3 + 5)$

   $f'(x) = (x^3+1)' \cdot (2x^3+5) + (x^3+1) \cdot (2x^3+5)'$

   $= 3x^2 \cdot (2x^3+5) + (x^3+1) \cdot (6x^2)$

2. The per-day cost function of the manufacture of portable MP3 players is given by

$$C(x) = 128{,}000 + 30x + x^3$$

where x is the number of MP3 players manufactured per day. Assume that the company cannot manufacture more than 100 MP3 players per day.

a. Find the marginal Cost function.

The marginal cost function is the derivative of the cost function.

$$MC = C'(x) = (128{,}000 + 30x + x^3)'$$

$$C'(x) = 30 + 3x^2$$

b. Interpret the meaning of C' (80).

$$C'(80) = 30 + 3 \times (80)^2 = 19{,}230$$

$C'(80) \approx$ the cost of producing the 81st item.

c. Write the average cost function

$$\text{Average cost} = \frac{\text{cost of producing } x \text{ items}}{x}$$

$$f(x) = \frac{C(x)}{x}$$

$$\Rightarrow f(x) = \frac{128{,}000 + 30x + x^3}{x}$$

d. How many MP3 players should be manufactured in order to minimize the average cost?

we need to find x to minimize $f(x)$

$$f(x) = \frac{128{,}000 + 30x + x^3}{x}$$

$$f(x) = \frac{128{,}000}{x} + \frac{30x}{x} + \frac{x^3}{x}$$

$$f(x) = 128000 \cdot x^{-1} + 30 + x^2$$

Step 1: $f'(x) = \left[128000 \cdot x^{-1} + 30 + x^2\right]'$

$$= 128{,}000 \cdot (-1) \cdot x^{-1-1} + 0 + 2 \cdot x^{2-1}$$

$$\boxed{f'(x) = -128{,}000 \cdot x^{-2} + 2x}$$

Step 2: Solve $f'(x) = 0$

$$-128{,}000\, x^{-2} + 2x = 0$$

$$\Rightarrow \quad 2x = 128{,}000 \cdot x^{-2}$$

$$\Rightarrow \quad 2x = \frac{128{,}000}{x^2}$$

$$2x \cdot x^2 = 128{,}000$$

$$2x^3 = 128{,}000$$

$$x^3 = \frac{128{,}000}{2} = 64000$$

$$x = \sqrt[3]{64000} = 40$$

Step 3: Confirm that $f''(40) > 0$

$$f''(x) = \left(-128{,}000\, x^{-2} + 2x\right)'$$

$$= -128{,}000 \cdot (-2) \cdot x^{-2-1} + 2$$

$$= 256\,000\, x^{-3} + 2$$

$$f''(x) = \frac{256000}{x^3} + 2$$

$$\Rightarrow \quad f''(40) = \frac{256{,}000}{40^3} + 2 > 0$$

$x = 40$ would minimize the average cost.

3. A luxury watch manufacturer has determined that the daily price p (in dollars) they can charge is related to the quantity x they produce by the following quadratic demand function.

$$p(x) = -x^2 + 120x + 2700$$

a. Find the revenue function and the marginal revenue function

(*) Revenue = Price * Quantity

$$R(x) = (-x^2 + 120x + 2700) * x$$

$$R(x) = -x^3 + 120x^2 + 2700x$$

(*) Marginal Revenue = $MR(x) = R'(x)$

$$\Rightarrow R'(x) = (-x^3 + 120x^2 + 2700x)'$$

M

$$\Rightarrow R'(x) = -3x^2 + 120 \cdot 2 \cdot x^{2-1} + 2700$$

$$R'(x) = -3x^2 + 240x + 2700$$

b. Interpret the meaning of MR (10).

$$MR(10) = R'(10) = -3 * 10^2 + 240 * 10 + 2700$$

$$= 4800$$

$MR(10) \approx$ Revenue of selling the 11th item

c. Find the quantity to maximize the revenue function

we need to find x to maximize $R(x)$

(*) Step 1: Find $R'(x)$

$$R'(x) = -3x^2 + 240x + 2700$$

(*) Step 2: Solve $R'(x) = 0$

$$-3x^2 + 240x + 2700 = 0$$

$$\Rightarrow \quad -3 \cdot (x^2 - 80x - 900) = 0$$

$$\Rightarrow \quad x^2 - 80x - 900 = 0$$

$$\Rightarrow \quad (x - 90) \cdot (x + 10) = 0$$

$$\Rightarrow \quad x = 90 \; ; \quad x = -10$$

Because x is the quantity, it has to be positive

$$\Rightarrow \quad \boxed{x = 90}$$

Step 3: confirm that $R''(90) < 0$

$$R'(x) = -3x^2 + 240x + 2700$$

$$R''(x) = -6x + 240$$

$R''(90) = -6 * 90 + 240 = -300 < 0$

This confirms that $R(x)$ is maximized at $x = 90$.

4. A company determines that when x units of a product are produced each month, they will be sold at the price of

$$p = 22.2 - 1.2x$$

dollars per unit. The total cost of producing the x units will be

$$C(x) = 0.4x^2 + 3x + 40$$

a. Find the profit function and the marginal profit function

(*) Profit = Revenue − Cost

$P(x)$ = price * quantity − cost

$= (22.2 - 1.2x)x - (0.4x^2 + 3x + 40)$

$P(x) = 22.2x - 1.2x^2 - .4x^2 - 3x - 40$

$$P(x) = -1.6x^2 + 19.2x - 40$$

(*) Marginal Profit

$$MP(x) = P'(x) = (-1.6x^2 + 19.2x - 40)$$

$$\Rightarrow \boxed{MP(x) = -3.2x + 19.2}$$

b. Interpret the meaning of MP(5).

$$MP(5) = -3.2 * 5 + 19.2 = 3.2$$

c. How many units should the company produce to maximize the profit?

Step 1: Find $P'(x)$

$$P'(x) = -3.2x + 19.2$$

Step 2: Solve $P'(x) = 0$

$$-3.2x + 19.2 = 0$$

$$x = \frac{19.2}{3.2} = 6$$

Step 3: confirm that $P''(6) < 0$

$$P''(x) = (-3.2x + 19.2)'$$

$$P''(x) = -3.2 < 0$$

$$P''(6) < 0$$

$\Rightarrow$ $P'(x)$ is maximized at $x = 6$